HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed, built, and tested clinical study databases within an Electronic Data Capture (EDC) system, adhering to strict quality standards and study timelines.
- Developed and configured electronic Case Report Forms (eCRFs) and implemented complex edit checks to ensure data integrity and compliance with clinical trial protocols.
- Supported the integration of clinical data transfers and performed database validation activities, collaborating closely with Data Management and project teams.
- Troubleshot and resolved technical issues within EDC systems, providing ongoing support for multiple clinical studies and progressively taking ownership of study builds.
- Contributed to process improvement initiatives, exploring automation opportunities and innovative solutions to enhance study development efficiency.
- Collaborated with sponsors and cross-functional teams to gather requirements, support User Acceptance Testing (UAT), and ensure successful study delivery from build to closeout.
- Containerized data processing scripts using **Docker** and deployed them to **AWS Lambda**, enabling scalable and efficient execution for ad-hoc data validation tasks.

SKILLS

EDC Systems – Medidata Rave, Electronic Data Capture (EDC)

Programming & Scripting – Python, SQL, Java, C#

Data Management – eCRFs, Edit Checks, Database Validation, Clinical Data Management

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Jenkins

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile, Scrum

PROJECTS

Clinical Study Build and Validation Automation

- Automated the generation of standard eCRF templates and edit check logic using **Python scripting**, reducing manual build time by an estimated **30%** for routine studies.
- Developed a reusable **SQL** framework for validating data consistency across multiple clinical study databases, flagging **~15%** of common data anomalies during pre-validation checks.
- Created **Docker** containers for data extraction and transformation scripts, ensuring consistent execution environments for database validation and integration tasks.

Data Integration Pipeline for Research Studies

- Designed and implemented a data pipeline using **Python** and **AWS S3** to ingest and stage data from various research sources for analysis.
- Built **REST APIs** using **Java Spring Boot** to expose processed clinical trial data to downstream analytics platforms, facilitating cross-system integration.
- Collaborated with the research analytics team to define data transformation rules and ensure output met their requirements for subsequent statistical analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)